Selection Games with Compact Sets

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- Our initial question: How much C_p stuff can be adapted for C_k?

Thematic question

If it can be done for points/finite sets, why not compact sets?

Context for compact sets

The Vietoris topology

Let $\mathbb{K}(X)$ denote the set of compact subsets of a topological space X endowed with the topology with the sets of the form

$$[U_1, U_2, \ldots, U_n] = \left\{ K \in \mathbb{K}(X) : \left[K \subseteq \bigcup_{j=1}^{n} U_j \right] \wedge \bigwedge_{j=1}^{n} (K \cap U_j \neq \emptyset) \right\}$$

where each $U_j \subseteq X$ is open.

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where each $U_j \subseteq X$ is open.

➢ Why? ◃

$\mathbb{K}(X)$ is "well-behaved" and many important topological properties of X are reflected in $\mathbb{K}(X)$.

(E.g.; second-countability, separability, first-countability, Hausdorffness, (complete) regularity, metrizability...)

It also gives a context in which we can coherently translate statements about finite sets to statements about compact sets:

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A *finite* union of *finite* sets is *finite*.

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<u>Original</u>		<u>Translation</u>
A *finite* union of *finite* sets is *finite*.	$\longrightarrow$	A *compact* union of *compact* sets is *compact*.

The latter is, formally, if $\mathbf{K} \subseteq \mathbb{K}(X)$ is compact, then $\bigcup \mathbf{K}$ is a compact subset of X.

The topological protagonists

For a space X,

- $\mathbb{K}(X)$ is as above.
- $X^{<\omega} = \bigsqcup_{n\in\omega} X^{n+1}$, the disjoint union of the finite Tychonoff products.
- $\mathcal{P}_{\text{fin}}(X)$ is the set of finite subsets of X, topologized as a subspace of $\mathbb{K}(X)$.

Object	Cover type
points, X	open covers, $\mathcal{O}_X$
finite sets, $\mathcal{P}_{\text{fin}}(X)$	ω-covers, Ω_X
compact sets, $\mathbb{K}(X)$	k-covers, $\mathcal{K}_X$

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We mention $X^{<\omega}$ since we can capture some results of the form "X is $\mathfrak{P}$ if and only if every finite power of X is $\mathfrak{Q}$" in terms of $\mathcal{P}_{\text{fin}}(X)$.

Selection Principles

Let $\mathcal{A}$ and $\mathcal{B}$ be sets. Then

$$\mathsf{S}_1(\mathcal{A},\mathcal{B}) \equiv (\forall A \in \mathcal{A}^\omega)\left(\exists B \in \prod_{n\in\omega} A_n\right)\ [\{B_n : n \in \omega\} \in \mathcal{B}]$$

and

$$\mathsf{S}_{\mathrm{fin}}(\mathcal{A},\mathcal{B}) \equiv (\forall A \in \mathcal{A}^\omega)\left(\exists B \in \prod_{n\in\omega} [A_n]^{<\aleph_0}\right)\ \left[\bigcup\{B_n : n \in \omega\} \in \mathcal{B}\right]$$

If $\mathcal{A}$ and $\mathcal{B}$ are class functions defined on the class of topological spaces,

$$X \models \mathsf{S}_*(\mathcal{A},\mathcal{B}) \quad \text{means} \quad X \models \mathsf{S}_*(\mathcal{A}_X,\mathcal{B}_X).$$

Selection Principle	Name
$\mathsf{S}_1(\mathcal{O},\mathcal{O})$	Rothberger
$\mathsf{S}_{\mathrm{fin}}(\mathcal{O},\mathcal{O})$	Menger
$\mathsf{S}_1(\Omega,\Omega)$	ω-Rothberger
$\mathsf{S}_{\mathrm{fin}}(\Omega,\Omega)$	ω-Menger
$\mathsf{S}_1(\mathcal{K},\mathcal{K})$	k-Rothberger
$\mathsf{S}_{\mathrm{fin}}(\mathcal{K},\mathcal{K})$	k-Menger

We will also discuss the *Hurewicz principle*: for every $\mathscr{U} \in \mathcal{O}_X^\omega$, there is $\mathscr{F} \in \prod_{n\in\omega} [\mathscr{U}_n]^{<\aleph_0}$ such that

$$(\forall x \in X)(\exists m \in \omega)(\forall n \geqslant m) \left[x \in \bigcup \mathscr{F}_n\right].$$

Groupability

Let $\mathfrak{C}$ be a class. Then $\mathscr{C} \in \mathfrak{C}$ is *groupable* if there is a finite-to-one $\varphi : \mathscr{C} \to \omega$ such that, for any $J \in [\omega]^{\aleph_0}$, $\left\{\varphi^{-1}(n) : n \in J\right\} \in \mathfrak{C}$.

$$\mathfrak{C}^{\mathrm{gp}} = \{\mathscr{C} \in \mathfrak{C} : \mathscr{C} \text{ is groupable}\}.$$

Some classical results

- (Gerlits-Nagy, 1982) Every finite power of X is Lindelöf if and only if X is an ϵ-space (ω-Lindelöf).
- (Sakai, 1988) Every finite power of X is Rothberger if and only if X is ω-Rothberger.
- (Just-Miller-Scheepers-Szeptycki, 1996) Every finite power of X is Menger if and only if X is ω-Menger.
- (Kočinac-Scheepers, 2003) Every finite power of X is Hurewicz if and only if $X \models \mathsf{S}_{\mathrm{fin}}(\Omega, \Omega^{\mathrm{gp}})$.
- (Kočinac-Scheepers, 2003) Every finite power of X is Gerlits-Nagy if and only if $X \models \mathsf{S}_1(\Omega, \Omega^{\mathrm{gp}})$.

A space is *Gerlits-Nagy* if it is both Hurewicz and Rothberger.

Observation concerning finite powers

By the above results, it is evident that each of the properties ω-Lindelöf, ω-Rothberger, ω-Menger, $\mathsf{S}_{\mathrm{fin}}(\Omega, \Omega^{\mathrm{gp}})$, and $\mathsf{S}_1(\Omega, \Omega^{\mathrm{gp}})$ are preserved under finite powers.

Can we do away with order and multiplicity?

The connection between ω-covers and finite powers should be clear, but perhaps keeping track of all tuples is unnecessary.

What about a compact analog?

The "for all finite powers of X" aspect doesn't seem to lend itself to a compact analog, but thinking in terms of a fixed hyperspace does.

The case for finite sets

Using $\mathcal{P}_{\text{fin}}(X)$ as our hyperspace, we obtain a natural replacement of the "all finite powers" criterion.

- (C.-Holshouser, 2022) $\mathcal{P}_{\text{fin}}(X)$ is Lindelöf if and only if X is an ϵ-space (ω-Lindelöf).
- (C.-Holshouser, 2022) $\mathcal{P}_{\text{fin}}(X)$ is Rothberger if and only if X is ω-Rothberger.
- (C.-Holshouser, 2022) $\mathcal{P}_{\text{fin}}(X)$ is Menger if and only if X is ω-Menger.
- (C., 2023) $\mathcal{P}_{\text{fin}}(X)$ is Hurewicz if and only if $X \models \mathsf{S}_{\text{fin}}(\Omega, \Omega^{\text{gp}})$.
- (C., 2023) $\mathcal{P}_{\text{fin}}(X)$ is Gerlits-Nagy if and only if $X \models \mathsf{S}_1(\Omega, \Omega^{\text{gp}})$.

The case for compact sets

- (C.-Holshouser, 2022) $\mathbb{K}(X)$ is Lindelöf if and only if X is k-Lindelöf.
- ~~$\mathbb{K}(X)$ is Rothberger if and only if X is k-Rothberger.~~
  - $\mathbb{R}$ witnesses the failure of this equivalence.
- (C.-Holshouser, 2022) $\mathbb{K}(X)$ is Menger if and only if X is k-Menger.
- (C., 2023) $\mathbb{K}(X)$ is Hurewicz if and only if $X \models \mathsf{S}_{\mathrm{fin}}(\mathcal{K}, \mathcal{K}^{\mathrm{gp}})$.
- ~~$\mathbb{K}(X)$ is Gerlits-Nagy if and only if $X \models \mathsf{S}_1(\mathcal{K}, \mathcal{K}^{\mathrm{gp}})$.~~
  - $\mathbb{R}$ witnesses the failure of this equivalence.

Games

Each of these selection principles have associated games of length ω with two players, P1 and P2.

For example, $\mathsf{G}_{\mathrm{fin}}(\mathcal{A}, \mathcal{B})$ is defined as:

- At round $n \in \omega$, P1 chooses $A_n \in \mathcal{A}$.
- P2 responds with $B_n \in [A_n]^{<\aleph_0}$.
- P2 wins if and only if $\bigcup\{B_n : n \in \omega\} \in \mathcal{B}$.

$\mathsf{G}_1(\mathcal{A}, \mathcal{B})$ has a similar definition and we'll use $\mathsf{HG}(X)$ to denote the Hurewicz game on X.

Strategy Types

- Constant (winning) strategies for P1: the non-existence of such a strategy is equivalent to a Lindelöf-like property.
- Predetermined (winning) strategies for P1: the non-existence of such a strategy is equivalent to the corresponding selection principle.
- (Full/Perfect-information) Winning strategies for P1.
- (Full/Perfect-information) Winning strategies for P2.
- Markov (winning) strategies for P2: these are typically seen as "traditional" topological properties.

Notation

- $\mathrm{I} \underset{\text{cnst}}{\uparrow} \mathcal{G}$ means P1 has a constant winning strategy in $\mathcal{G}$.
- $\mathrm{I} \underset{\text{pre}}{\uparrow} \mathcal{G}$ means P1 has a predetermined winning strategy in $\mathcal{G}$.
- $\mathrm{I} \uparrow \mathcal{G}$ means P1 has a winning strategy in $\mathcal{G}$.
- $\mathrm{II} \uparrow \mathcal{G}$ means P2 has a winning strategy in $\mathcal{G}$.
- $\mathrm{II} \underset{\text{mark}}{\uparrow} \mathcal{G}$ means P2 has a Markov winning strategy in $\mathcal{G}$.

Note

$$\mathrm{II} \underset{\text{mark}}{\uparrow} \mathcal{G} \Longrightarrow \mathrm{II} \uparrow \mathcal{G} \Longrightarrow \mathrm{I} \not\uparrow \mathcal{G} \Longrightarrow \mathrm{I} \underset{\text{pre}}{\not\uparrow} \mathcal{G} \Longrightarrow \mathrm{I} \underset{\text{cnst}}{\not\uparrow} \mathcal{G}$$

$$\mathrm{I} \underset{\text{pre}}{\not\uparrow} \mathcal{G} \Longleftrightarrow \text{Selection Principle}$$

$$\mathrm{I} \underset{\text{cnst}}{\not\uparrow} \mathcal{G} \Longleftrightarrow \text{Flavor of Lindelöf}$$

Classical P1 strategy reduction

- (Hurewicz, 1925)

$$\mathrm{I} \uparrow \mathsf{G}_{\mathrm{fin}}(\mathcal{O}_X, \mathcal{O}_X) \iff \mathrm{I} \underset{\mathrm{pre}}{\uparrow} \mathsf{G}_{\mathrm{fin}}(\mathcal{O}_X, \mathcal{O}_X).$$

- (Pawlikowski, 1994)

$$\mathrm{I} \uparrow \mathsf{G}_1(\mathcal{O}_X, \mathcal{O}_X) \iff \mathrm{I} \underset{\mathrm{pre}}{\uparrow} \mathsf{G}_1(\mathcal{O}_X, \mathcal{O}_X).$$

- (Scheepers, 1997)

$$\mathrm{I} \uparrow \mathsf{G}_*(\Omega_X, \Omega_X) \iff \mathrm{I} \underset{\mathrm{pre}}{\uparrow} \mathsf{G}_*(\Omega_X, \Omega_X).$$

- (Scheepers, 1996)

$$\mathrm{I} \uparrow \mathsf{HG}(X) \iff \mathrm{I} \underset{\mathrm{pre}}{\uparrow} \mathsf{HG}(X).$$

- (Kočinac-Scheepers, 2003)

$$\mathrm{I} \uparrow \mathsf{G}_*(\Omega_X, \Omega_X^{\mathrm{gp}}) \iff \mathrm{I} \underset{\mathrm{pre}}{\uparrow} \mathsf{G}_*(\Omega_X, \Omega_X^{\mathrm{gp}}).$$

P1 strategy reduction for $\mathcal{K}$

- (C.-Holshouser, 2022)

$$\mathrm{I} \uparrow \mathsf{G}_*(\mathcal{K}_X, \mathcal{K}_X) \iff \mathrm{I} \underset{\text{pre}}{\uparrow} \mathsf{G}_*(\mathcal{K}_X, \mathcal{K}_X).$$

- (C., 2023)

$$\mathrm{I} \uparrow \mathsf{G}_*(\mathcal{K}_X, \mathcal{K}_X^{\mathrm{gp}}) \iff \mathrm{I} \underset{\text{pre}}{\uparrow} \mathsf{G}_*(\mathcal{K}_X, \mathcal{K}_X^{\mathrm{gp}}).$$

Markov for P2 is "traditional"?

- For a T_1 space X,

$$X \text{ is countable} \iff \mathrm{II} \underset{\text{mark}}{\uparrow} \mathsf{G}_1(\Omega_X, \Omega_X) \iff \mathrm{II} \underset{\text{mark}}{\uparrow} \mathsf{G}_1(\mathcal{O}_X, \mathcal{O}_X).$$

- For a regular space X,

$$X \text{ is } \sigma\text{-compact} \iff \mathrm{II} \underset{\text{mark}}{\uparrow} \mathsf{G}_{\mathrm{fin}}(\Omega_X, \Omega_X) \iff \mathrm{II} \underset{\text{mark}}{\uparrow} \mathsf{G}_{\mathrm{fin}}(\mathcal{O}_X, \mathcal{O}_X).$$

- For any space X,

$$X \text{ is hemicompact} \iff \mathrm{II} \underset{\text{mark}}{\uparrow} \mathsf{G}_1(\mathcal{K}_X, \mathcal{K}_X).$$

If X is regular,

$$X \text{ is hemicompact} \iff \mathrm{II} \underset{\text{mark}}{\uparrow} \mathsf{G}_1(\mathcal{K}_X, \mathcal{K}_X) \iff \mathrm{II} \underset{\text{mark}}{\uparrow} \mathsf{G}_{\mathrm{fin}}(\mathcal{K}_X, \mathcal{K}_X).$$

Another observation related to products

Note that each of the properties of being countable, σ-compact, and hemicompact are not only preserved by finite powers, but are finitely productive.

Some productivity theorems

We use the adjective *strategically* to mean P2 has a winning strategy in the corresponding game.

- (Dias-Scheepers, 2015) The properties of being strategically Rothberger and strategically Menger are both finitely productive.
- (Clontz-Holshouser, 2019) The property of being strategically ω-Rothberger is equivalent to being strategically Rothberger, so the finite productivity extends.
- (Clontz, 2019) The property of being strategically ω-Menger is equivalent to being strategically Menger, so the finite productivity extends.

Productivity for the k-variants

- (C.-Clontz-Holshouser, 2024) The properties of being strategically k-Rothberger and strategically k-Menger are both finitely productive.

Some ZFC relationships

$$
\begin{tikzcd}
\text{Markov Rothberger} \arrow[r, "\not\Leftarrow"] \arrow[d, leftrightarrow] & \text{Strategically Rothberger} \arrow[r] \arrow[d, leftrightarrow] & \text{Rothberger} \\
\text{Markov } \omega\text{-Rothberger} \arrow[r, "\not\Leftarrow"] \arrow[d, "\not\Uparrow"] & \text{Strategically } \omega\text{-Rothberger} \arrow[r] \arrow[d, "\not\Uparrow"] & \omega\text{-Rothberger} \arrow[u] \arrow[d, "\not\Uparrow"] \\
\text{Markov } \omega\text{-Menger} \arrow[r, "\not\Leftarrow"] \arrow[d, leftrightarrow] & \text{Strategically } \omega\text{-Menger} \arrow[r] \arrow[d, leftrightarrow] & \omega\text{-Menger} \arrow[r, "\not\Leftarrow"] \arrow[d] & \omega\text{-Lindelöf} \arrow[d, "\not\Uparrow"] \\
\text{Markov Menger} \arrow[r, "\not\Leftarrow"] & \text{Strategically Menger} \arrow[r] & \text{Menger} \arrow[r, "\not\Leftarrow"] & \text{Lindelöf}
\end{tikzcd}
$$

More ZFC relationships

$$
\begin{tikzcd}
\text{Markov } k\text{-Rothberger} \arrow[r, "\not\Leftarrow"] \arrow[d] & \text{Strategically } k\text{-Rothberger} \arrow[r] \arrow[d] & k\text{-Rothberger} \arrow[d] & \\
\text{Markov } k\text{-Menger} \arrow[r, "\not\Leftarrow"] \arrow[d, "\not\Uparrow"] & \text{Strategically } k\text{-Menger} \arrow[r] \arrow[d, "\not\Uparrow"] & k\text{-Menger} \arrow[r, "\not\Leftarrow"] \arrow[d, "\not\Uparrow"] & k\text{-Lindelöf} \arrow[d] \\
\text{Markov } \omega\text{-Menger} \arrow[r, "\not\Leftarrow"] & \text{Strategically } \omega\text{-Menger} \arrow[r] & \omega\text{-Menger} \arrow[r, "\not\Leftarrow"] & \omega\text{-Lindelöf} \\
\text{Markov } \omega\text{-Rothberger} \arrow[r, "\not\Leftarrow"] \arrow[u, "\not\Downarrow"] & \text{Strategically } \omega\text{-Rothberger} \arrow[r] \arrow[u, "\not\Downarrow"] & \omega\text{-Rothberger} \arrow[u] &
\end{tikzcd}
$$

More ZFC relationships

$$
\begin{tikzcd}
\text{Markov } k\text{-Rothberger} \arrow[r, "\not\Leftarrow"] \arrow[d, "!!"] & \text{Strategically } k\text{-Rothberger} \arrow[r] \arrow[d] & k\text{-Rothberger} \arrow[d] & \\
\text{Markov } k\text{-Menger} \arrow[r, "\not\Leftarrow"] \arrow[d, "\not\Uparrow"] & \text{Strategically } k\text{-Menger} \arrow[r] \arrow[d, "\not\Uparrow"] & k\text{-Menger} \arrow[r, "\not\Leftarrow"] \arrow[d, "\not\Uparrow"] & k\text{-Lindelöf} \arrow[d, "!!"] \\
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\text{Markov } \omega\text{-Rothberger} \arrow[r, "\not\Leftarrow"] \arrow[u, "\not\Downarrow"] & \text{Strategically } \omega\text{-Rothberger} \arrow[r] \arrow[u, "\not\Downarrow"] & \omega\text{-Rothberger} \arrow[u] &
\end{tikzcd}
$$

The role of game equivalence

Nearly all of the results above concerning $\mathcal{P}_{\mathrm{fin}}(X)$ and $\mathbb{K}(X)$ follow from general game equivalences.

- The relationship $\mathcal{G} \equiv \mathcal{H}$ between two games means that $\mathcal{G}$ and $\mathcal{H}$ agree on the existence of each strategy type except potentially constant strategies for P1 (this will be relevant for the case for continuous functions).

- The relationship $\mathcal{G} \rightleftarrows \mathcal{H}$ between two games means that both games agree on the existence of each strategy type, including constant strategies for P1.

Some equivalent games

Theorem (C.-Holshouser, 2022)

For any space X,

$$\mathsf{G}_*(\Omega_X, \Omega_X) \rightleftarrows \mathsf{G}_*(\Omega_{X^{<\omega}}, \Omega_{X^{<\omega}}) \rightleftarrows \mathsf{G}_*(\mathcal{O}_{X^{<\omega}}, \mathcal{O}_{X^{<\omega}})$$
$$\rightleftarrows \mathsf{G}_*(\Omega_{\mathcal{P}_{\mathrm{fin}}(X)}, \Omega_{\mathcal{P}_{\mathrm{fin}}(X)}) \rightleftarrows \mathsf{G}_*(\mathcal{O}_{\mathcal{P}_{\mathrm{fin}}(X)}, \mathcal{O}_{\mathcal{P}_{\mathrm{fin}}(X)})$$

and

$$\mathsf{G}_{\mathrm{fin}}(\mathcal{K}_X, \mathcal{K}_X) \rightleftarrows \mathsf{G}_{\mathrm{fin}}(\mathcal{K}_{\mathbb{K}(X)}, \mathcal{K}_{\mathbb{K}(X)}) \rightleftarrows \mathsf{G}_{\mathrm{fin}}(\mathcal{O}_{\mathbb{K}(X)}, \mathcal{O}_{\mathbb{K}(X)}).$$

- $\mathbb{R}$ witnesses the impossibility of an equivalence at the single-selection level for the k-variant.

Core idea for the k-Menger game

For games $\mathcal{G}$ and $\mathcal{H}$, we write $\mathcal{G} \leqslant_{\mathrm{II}} \mathcal{H}$ to mean

$$\begin{cases} \mathrm{II} \underset{\text{mark}}{\uparrow} \mathcal{G} & \Longrightarrow & \mathrm{II} \underset{\text{mark}}{\uparrow} \mathcal{H} \\ \mathrm{II} \uparrow \mathcal{G} & \Longrightarrow & \mathrm{II} \uparrow \mathcal{H} \\ \mathrm{I} \not\uparrow \mathcal{G} & \Longrightarrow & \mathrm{I} \not\uparrow \mathcal{H} \\ \mathrm{I} \underset{\text{pre}}{\not\uparrow} \mathcal{G} & \Longrightarrow & \mathrm{I} \underset{\text{pre}}{\not\uparrow} \mathcal{H} \end{cases}$$

If, in addition,

$$\mathrm{I} \underset{\text{cnst}}{\not\uparrow} \mathcal{G} \Longrightarrow \mathrm{I} \underset{\text{cnst}}{\not\uparrow} \mathcal{H},$$

then we write $\mathcal{G} \leqslant_{\mathrm{II}}^{+} \mathcal{H}$.

$\mathsf{G}_{\mathrm{fin}}(\mathcal{K}_Y, \mathcal{K}_Y) \leqslant^+_{\mathrm{II}} \mathsf{G}_{\mathrm{fin}}(\mathcal{O}_Y, \mathcal{O}_Y)$

For any open cover $\mathscr{U}_n$ of Y,

$$\mathscr{V}_n = \left\{ \bigcup \mathscr{F} : \mathscr{F} \in [\mathscr{U}_n]^{<\aleph_0} \right\}$$

is a k-cover of Y. Any sequence of finite selections from each $\mathscr{V}_n$ resulting in a k-cover of Y corresponds to a finite selection from each $\mathscr{U}_n$ which results in an open cover of Y.

$\mathsf{G}_{\mathrm{fin}}(\mathcal{K}_X, \mathcal{K}_X) \leftrightarrows \mathsf{G}_{\mathrm{fin}}(\mathcal{K}_{\mathbb{K}(X)}, \mathcal{K}_{\mathbb{K}(X)})$

This equivalence boils down to a correspondence between k-covers of X and k-covers of $\mathbb{K}(X)$.

- If $\mathscr{U}$ is a k-cover of X, then $\{[U] : U \in \mathscr{U}\}$ is a k-cover of $\mathbb{K}(X)$.
  - For any $\mathbf{K} \subseteq \mathbb{K}(X)$ which is compact, $\bigcup \mathbf{K} \subseteq X$ is compact. Then there is some $U \in \mathscr{U}$ for which $\bigcup \mathbf{K} \subseteq U$. Note then that $\mathbf{K} \subseteq [U]$.
- If $\mathscr{U}$ is a k-cover of $\mathbb{K}(X)$, then

$$\{V \in \mathscr{T}_X : (\exists U \in \mathscr{U})\ [V] \subseteq U\}$$

is a k-cover of X.

- Let $K \subseteq X$ be compact and note that $\mathbf{K} := \{F \in \mathbb{K}(X) : F \subseteq K\}$ is a compact subset of $\mathbb{K}(X)$.
- Let $U \in \mathscr{U}$ be such that $\mathbf{K} \subseteq U$ and, for each $F \in \mathbf{K}$, let B_F be a basic neighborhood s.t. $F \in B_F \subseteq U$.
- By compactness, let $F_1, \ldots, F_n \in \mathbf{K}$ be such that $\mathbf{K} \subseteq \bigcup_{j=1}^{n} B_{F_j}$.
- Let $B_{F_j} = [W_{j,1}, W_{j,2}, \ldots, W_{j,m_j}]$ and, for $x \in K$, $N_x = \bigcap\{W_{j,k} : x \in W_{j,k}\}$.
- $V = \bigcup_{x \in K} N_x$.
- If $L \in [V]$, $L \subseteq \bigcup_{x \in K} N_x$ and there is some $\{x_1, x_2, \ldots, x_p\} \subseteq K$ such that $L \subseteq \bigcup_{j=1}^{p} N_{x_j}$. We can also guarantee that $L \cap N_{x_j} \neq \emptyset$ for each $j = 1, 2, \ldots, p$.
- $\{x_1, x_2, \ldots, x_p\} \in \mathbf{K}$ so it is contained in some $[W_{j,1}, W_{j,2}, \ldots, W_{j,m_j}]$.
- Then $L \in [W_{j,1}, W_{j,2}, \ldots, W_{j,m_j}] \subseteq U$. So $[V] \subseteq U$.

$\mathsf{G}_{\mathrm{fin}}(\mathcal{O}_{\mathbb{K}(X)}, \mathcal{O}_{\mathbb{K}(X)}) \leqslant^{+}_{\mathrm{II}} \mathsf{G}_{\mathrm{fin}}(\mathcal{K}_X, \mathcal{K}_X)$

- Every k-cover $\mathscr{U}_n$ of X is translated to

$$\mathscr{V}_n = \{[U] : U \in \mathscr{U}_n\},$$

  an open cover of $\mathbb{K}(X)$.
- Every finite subset of $\mathscr{V}_n$ corresponds to a finite subset of $\mathscr{U}_n$.
- If the resulting selection from each $\mathscr{V}_n$ is an open cover $\mathbb{K}(X)$, then, for $K \subseteq X$ compact, there is some single choice $[U]$ for which $K \in [U] \implies K \subseteq U$.
- Hence, the resulting selection from $\mathscr{U}_n$ forms a k-cover of X.

Connections with C_p/C_k

For a Tychonoff space X, $C_p(X)$ denotes the set of continuous real-valued functions on X with the pointwise convergence topology and $C_k(X)$ denotes the set of continuous real-valued functions on X endowed with the topology of uniform convergence on compacta.

Our relationships with the properties above require some supporting characters.

- For a space Y, $\mathscr{D}_Y$ is the set of all dense subsets of Y.
- For a space Y and $y \in Y$, $\Omega_{Y,y}$ is the set of all subsets of Y containing y in their closure.
- For a space Y and $y \in Y$, $\mathscr{N}_{Y,y}$ is the filter of open neighborhoods of y.

More equivalences...

Theorem

For a Tychonoff space X,

$$\mathsf{G}_1(\Omega_X, \Omega_X) \equiv \mathsf{G}_1(\Omega_{C_p(X),\mathbf{0}}, \Omega_{C_p(X),\mathbf{0}}) \equiv \mathsf{G}_1(\mathscr{D}_{C_p(X),\mathbf{0}}, \Omega_{C_p(X),\mathbf{0}}),$$

$$\mathsf{G}_1(\mathscr{N}_{C_p(X),\mathbf{0}}, \neg\Omega_{C_p(X),\mathbf{0}}) \equiv \mathsf{G}_1(\mathscr{T}_{C_p(X),\mathbf{0}}, \neg\Omega_{C_p(X),\mathbf{0}}),$$

the two groups are dual, and the latter group is equivalent to a modification of the point-open game.

Note that we have used $\equiv$ here instead of $\leftrightarrows$ since keeping track of the turn number is necessary to ensure clustering at the constant zero function $\mathbf{0}$.

It works for k, too.

Theorem (C.-Holshouser, 2020)

For a Tychonoff space X,

$$\mathsf{G}_1(\mathcal{K}_X, \mathcal{K}_X) \equiv \mathsf{G}_1(\Omega_{C_k(X),\mathbf{0}}, \Omega_{C_k(X),\mathbf{0}}) \equiv \mathsf{G}_1(\mathscr{D}_{C_k(X),\mathbf{0}}, \Omega_{C_k(X),\mathbf{0}}),$$

$$\mathsf{G}_1(\mathscr{N}_{C_k(X),\mathbf{0}}, \neg\Omega_{C_k(X),\mathbf{0}}) \equiv \mathsf{G}_1(\mathscr{T}_{C_k(X),\mathbf{0}}, \neg\Omega_{C_k(X),\mathbf{0}}),$$

the two groups are dual, and the latter group is equivalent to a modification of the point-open game.

As some particular corollaries.

- The following are equivalent:
  - X is hemicompact.
  - $\mathbb{K}(X)$ is hemicompact.
  - $\mathbb{K}(X)$ is σ-compact.
  - $C_k(X)$ is metrizble.
- The following are equivalent:
  - X is k-Rothberger.
  - $C_k(X)$ has strong countable fan-tightness at $\mathbf{0}$.
  - $C_k(X)$ has strong countable dense fan-tightness at $\mathbf{0}$.

Incorporating the groupability condition, we can tie together some selection principles.

Theorem

- $\mathcal{P}_{\text{fin}}(X)$ is Hurewicz if and only if

$$C_p(X) \models \mathsf{S}_{\text{fin}}(\Omega_{C_p(X),\mathbf{0}}, \Omega^{\text{gp}}_{C_p(X),\mathbf{0}}).$$

- $\mathcal{P}_{\text{fin}}(X)$ is Gerlits-Nagy if and only if

$$C_p(X) \models \mathsf{S}_1(\Omega_{C_p(X),\mathbf{0}}, \Omega^{\text{gp}}_{C_p(X),\mathbf{0}}).$$

The version of this theorem in terms of only X and $C_p(X)$ was known already to Kočinac and Scheepers in 2002.

Theorem (C., 2023)

- $\mathbb{K}(X)$ is Hurewicz if and only if

$$C_k(X) \models \mathsf{S}_{\mathrm{fin}}(\Omega_{C_k(X),\mathbf{0}}, \Omega^{\mathrm{gp}}_{C_k(X),\mathbf{0}}).$$

- $X \models \mathsf{S}_1(\mathcal{K}, \mathcal{K}^{\mathrm{gp}})$ if and only if

$$C_k(X) \models \mathsf{S}_1(\Omega_{C_k(X),\mathbf{0}}, \Omega^{\mathrm{gp}}_{C_k(X),\mathbf{0}}).$$

Kočinac proved one direction of these equivalences (not through $\mathbb{K}(X)$) in 2003, but left the converses as an open question.

Thank You!